



Handheld UAV Detection and Positioning Device

Technical Specifications v1.0



Product Name:

Handheld Drone Detection and Positioning Device

Model name: ST-2421



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一、 Product Overview

H2L is a portable product that integrates drone spectrum detection, protocol analysis, RID identification and other regulatory functions. The single machine can realize real-time detection and early warning of drones, and accurate positioning and trajectory tracking of common drones and pilots (remote controllers).

The product has a compact appearance, is easy to carry, and has long battery life. It can be used for low-altitude security operations in scenes such as major event security, guard mission support, public security patrol, special counter-terrorism, political core areas, border defense, airports, power and petrochemical parks, etc.

二、 Product Composition

The product is mainly composed of a host, a detection antenna, a Type-C interface charger, and a protective box, as shown in the following table:

Table 1 Product hardware components list

No.	Device Name	Qty	Unit	Notes
1	Detection host	1	set	
2	Detection antenna	2	piece	
3	Type-c interface charger	1	piece	
4	Protective box	1	piece	

三、 Product Features

1) Drone positioning[1]: A single machine can locate the position of a drone, which can be displayed as a "drone icon" on the electronic map interface.

2) Pilot positioning[2]: A single machine can locate the position of the remote controller (pilot), which can be displayed as a "remote controller icon" on the electronic map interface.

3) Detection and identification[3]: It has a spectrum detection function and can identify common models such as DJI and Autel, as well as a variety of models such as homemade cross-country drones and WiFi drones.

4) FPV detection: It can detect and identify homemade cross-country



drones and analyze their first-person flight images in real time.

5) Pilot position navigation[4]: The system can plan the action route of the drone operator and provide a "one-click position navigation" function.

6) Unique identity identification[5]: It can identify the unique serial number (SN) of DJI series drones (using OcuSync), RID protocol drones, and WIFI drones.

7) Tracking[6]: It can display the flight tracks of multiple drones simultaneously on the electronic map interface.

8) Multi-dimensional information monitoring [7]: The model, location information (latitude and longitude), speed, altitude and pilot location (latitude and longitude) of the current drone can be displayed in real time in the detection list column.

9) Black and white list [8]: A white list can be set, and the system will not alarm when it detects a drone in the white list.

10) RID monitoring [9]: It has RID remote identification function and supports multiple standard protocols.

11) Intrusion alarm: When a drone intrusion is detected, the device will alarm by sound, vibration, etc.

12) Trajectory playback: Supports historical trajectory playback function to assist security personnel in analyzing historical drone flight data.

13) Detection record: The detection record list can retain historical detection records, including multi-dimensional information such as drone serial number, model, frequency, etc.

14) Visual data analysis: Supports drone detection and flight data statistical analysis, which can be displayed through a graphical interface.

15) Network intercom: The device can connect to a wireless network and support real-time intercom function between devices. (Optional function)

Note: The locatable drones in [1]-[8] are DJI models that use OcuSync image transmission.

[9] GB 42590-2023, ASTM F3411 Remote ID, ASD-STAN prEN 4709-002



Features

- 1) Passive detection:** no electromagnetic environmental pollution, environmentally friendly, high concealment.
- 2) Large screen touch interaction:** adopts a 6-inch large screen touch design, which is simple, convenient and fast to operate.
- 3) Portable:** the device weighs only 650g, can be easily carried and used immediately.
- 4) Standard charging interface:** Type-c standard charging interface, can be charged with an external power adapter or mobile power supply.

Product index parameters

5.1 Performance Indicators

Table 2 Product performance indicators

No.	Performance Indicators	Indicator parameters	Notes
1	Detection frequency	100MHz ~ 6GHz	
2	Detection Range	1.5~2km (Urban Environment) 2~3km (Open environment)	
3	Response time	3~5 s	
4	Position Range	1~2km (Urban Environment) 2~3km (Open environment)	
5	Detection quantity	≥10 flights (simultaneous)	
6	Battery life	3~4 h	



5.2 Mechanical parameters

Table 3 Mechanical parameters

No.	Index	Parameters	Notes
1	Host size	L*W*H: (185mm*80mm*33mm) ±5mm	
2	weight	(650±10) g	No antenna included

5.3 Electrical Characteristics

Table 4 Electrical characteristics

No.	Index	Parameters	Notes
1	Battery capacity	8000mAh	Built-in lithium battery
2	Battery life	4~6 hours	Screen brightness is 20%, room temperature is 25°C
3	Charging time	~ 3 hours	Use the original charger

5.4 Environmental adaptability

Table 5 Environmental adaptability

No.	Index	Parameters	Notes
1	Operating temperature	-25~40°C	
2	Storage temperature	-35~40°C	



5.5 Touch screen parameters

Table 6 Touch screen parameters

No.	Index	Parameters	Notes
1	Screen size	6 inches	
2	Screen resolution	1080*2160 pixels	

5.6 System Storage

Table 7 System storage parameters

No.	Index	Parameters	Notes
1	Memory	8GB	
2	Storage	256GB	

5.7 Operating Environment

Table 8 System operating parameters

No.	Index	Parameters	Notes
1	Operating System	Android 12	
2	Network Environment	Support Internet and stand-alone offline mode	